

Cloud & SaaS Integrations

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Extending Visibility Beyond OneAgent

While OneAgent provides deep application and infrastructure monitoring, many organizations need visibility into cloud services and SaaS platforms that can't run an agent. This notebook covers how to integrate AWS, Azure, GCP, and third-party SaaS tools into Dynatrace.

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Prerequisites

- Dynatrace environment with admin access
- **ActiveGate deployed** (required for Extensions 2.0 and classic cloud integrations; not needed if using the Clouds app)
- Cloud provider admin access (for AWS/Azure/GCP)

- API credentials for SaaS platforms

1. Integration Overview

Dynatrace offers multiple integration methods depending on the data source:

Method	Use Case	Requires ActiveGate?
Cloud Integrations (Classic)	AWS, Azure, GCP native services	Yes (for polling)
Clouds App (Modern)	AWS (GA), Azure (preview), GCP (preview) direct connections	No
Extensions 2.0	Custom data sources, SaaS APIs	Yes
OpenTelemetry	OTel-instrumented apps	No (direct ingest)
Log Ingest	External log sources	Optional
Metrics Ingest	Custom metrics via API	No

Why Integrate Before OneAgent?

Benefit	Description
Infrastructure context	See cloud resources before deploying agents
Dependency mapping	Understand managed services (RDS, Lambda, etc.)
Cost visibility	Cloud cost data available immediately
Complete topology	Smartscape includes cloud services from day one

2. AWS Integration

The AWS integration pulls metrics from CloudWatch and discovers AWS resources.

Supported Services

Category	Services
Compute	EC2, Lambda, ECS, EKS, Fargate
Database	RDS, DynamoDB, ElastiCache, DocumentDB
Storage	S3, EBS, EFS
Networking	ELB, ALB, NLB, API Gateway, CloudFront
Messaging	SQS, SNS, Kinesis, MSK
Other	Step Functions, Secrets Manager, and more

Setup Methods

Method	Best For	Complexity
IAM Role (recommended)	Production, multi-account	Medium
Access Key	Quick testing	Low
CloudFormation	Automated setup	Low

IAM Role Setup

1. Create an IAM role with CloudWatch read permissions
2. Add trust relationship for Dynatrace's AWS account
3. Configure in Dynatrace: Settings → Cloud and virtualization → AWS

Required IAM Policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "cloudwatch:GetMetricData",
        "cloudwatch:GetMetricStatistics",
        "cloudwatch:ListMetrics",
        "tag:GetResources",
        "tag:GetTagKeys",
        "tag:GetTagValues",
        "ec2:DescribeInstances",
        "ec2:DescribeVolumes",
        "rds:DescribeDBInstances",
        "lambda:ListFunctions",
        "lambda:GetFunction"
      ],
      "Resource": "*"
    }
  ]
}
```

Configuration Location

Path: Settings → Cloud and virtualization → AWS

Setting	Recommendation
Polling interval	5 minutes (default)
Services to monitor	Start with core services, expand as needed
Regions	Only regions where you have resources
Resource tags	Use tags to filter monitored resources

3. Azure Integration

The Azure integration uses Azure Monitor to collect metrics and discover resources.

Supported Services

Category	Services
Compute	Virtual Machines, App Service, Functions, AKS
Database	SQL Database, Cosmos DB, Redis Cache
Storage	Blob, Files, Queues, Tables
Networking	Load Balancer, Application Gateway, VNet
Messaging	Service Bus, Event Hubs, Event Grid

Setup via App Registration

1. Create an App Registration in Azure AD
2. Grant **Reader** role on subscriptions to monitor
3. Create a client secret
4. Configure in Dynatrace with:
 - Tenant ID
 - Client ID
 - Client Secret
 - Subscription IDs

Configuration Location: Settings → Cloud and virtualization → Azure

4. GCP Integration

The GCP integration uses Cloud Monitoring (formerly Stackdriver) APIs.

Supported Services

Category	Services
Compute	Compute Engine, GKE, Cloud Run, Cloud Functions
Database	Cloud SQL, Cloud Spanner, Firestore, Bigtable
Storage	Cloud Storage

Category	Services
Networking	Load Balancing, Cloud CDN
Messaging	Pub/Sub

Setup via Service Account

1. Create a Service Account in GCP
2. Grant **Monitoring Viewer** role
3. Generate and download JSON key
4. Configure in Dynatrace

Configuration Location: Settings → Cloud and virtualization → Google Cloud Platform

5. Extensions 2.0 Framework

Extensions 2.0 is Dynatrace's framework for integrating any data source - databases, SaaS platforms, network devices, and more.

How Extensions Work

Component	Role
Extension Package	Python code + metadata defining what to collect
ActiveGate	Executes extension code, polls data sources
Monitoring Configuration	Instance-specific settings (endpoints, credentials)
Dynatrace	Receives metrics, logs, events from extension

Extension Types

Type	Examples
Database	Oracle, SQL Server, PostgreSQL, MongoDB
Infrastructure	VMware, SNMP devices, F5, NetApp

Type	Examples
SaaS	Salesforce, ServiceNow, Jira, Confluent
Custom	Any REST API, custom protocols

Installing an Extension

1. **Find the extension** in Dynatrace Hub
2. **Install** to your environment
3. **Configure** with endpoint and credentials
4. **Assign** to an ActiveGate group
5. **Verify** data is flowing

Extension Configuration Example

```
# Example: Generic REST API extension configuration
enabled: true
description: "My SaaS API"
endpoints:
  - url: "https://api.example.com/metrics"
    authentication:
      type: "bearer"
      token: "${SAAS_API_TOKEN}"
      polling_interval: 60
activeGate:
  group: "saas-integrations"
```

Creating Custom Extensions

For SaaS platforms without a pre-built extension, you can create custom extensions:

1. Use the Extensions 2.0 SDK
2. Define metrics schema in YAML
3. Write Python code to fetch data
4. Package and upload to Dynatrace

SDK Documentation: [Extensions 2.0 Development](#)

6. Dynatrace Hub

The Dynatrace Hub is your marketplace for extensions, integrations, and apps.

Accessing the Hub

Path: Apps → Dynatrace Hub

Or use quick search: **Cmd+K** → "Hub"

Hub Categories

Category	Examples
Cloud	AWS, Azure, GCP, Kubernetes
Database	Oracle, SQL Server, PostgreSQL
Infrastructure	VMware, SNMP, NetApp
Messaging	Kafka, RabbitMQ, IBM MQ
Observability	OpenTelemetry, Prometheus
Security	Snyk, SonarQube
ITSM	ServiceNow, Jira, PagerDuty

Installing from Hub

1. Search for the integration you need
2. Click **Install** or **Add to environment**
3. Follow the configuration wizard
4. Verify data appears in Dynatrace

7. Common SaaS Integrations

Messaging Platforms

Platform	Integration Method	Key Metrics
Confluent Cloud	Extensions 2.0	Consumer lag, throughput, partitions
Amazon MSK	AWS Integration	Broker metrics, topic metrics
RabbitMQ	Extensions 2.0	Queue depth, message rates

CRM & Business Apps

Platform	Integration Method	Key Metrics
Salesforce	Extensions 2.0 / API	API calls, response times, limits
ServiceNow	Extensions 2.0	Incident metrics, workflow times

DevOps Tools

Platform	Integration Method	Key Metrics
GitHub	Webhooks	Deployment events, commit info
GitLab	Webhooks	Pipeline events, deployments
ArgoCD	Extensions 2.0	Sync status, app health

Database-as-a-Service

Platform	Integration Method	Key Metrics
MongoDB Atlas	Extensions 2.0	Connections, ops/sec, replication lag
AWS DocumentDB	AWS Integration	CloudWatch metrics
Snowflake	Extensions 2.0	Query performance, warehouse usage

8. Verifying Integrations

After configuring integrations, verify data is flowing into Dynatrace.

```
// Check for AWS entities
fetch dt.entity.aws_lambda_function
| fields entity.name, id
| limit 20

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes AWS_LAMBDA_FUNCTION
// | fields name, id
// | limit 20
```

```
// Check for Azure entities
fetch dt.entity.azure_vm
| fields entity.name, id
| limit 20
```

```
// Check for cloud-sourced metrics
// Note: There is no "metrics" starting command in DQL.
// Use timeseries to query a specific cloud metric, for example:
timeseries avg(dt.cloud.aws.lambda.duration), from:-24h
// Or browse available cloud metrics in the Dynatrace Metrics browser:
// Observe & Explore → Metrics → search for "cloud" or "aws" / "azure" /
"gcp"
```

```
// Check Extensions 2.0 status
// Note: "dt.entity.extensions:extension" is not a valid entity type for DQL
fetch.
// Use the Extensions API v2 instead:
// GET /api/v2/extensions
// Or check installed extensions in the Dynatrace Hub:
// Apps → Dynatrace Hub → Installed
// To list extension-reported entities, query a known extension entity type:
fetch dt.entity.host
| fieldsKeep entity.name, id
| limit 10

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes HOST
// | fieldsKeep name, id
// | limit 10
```

Troubleshooting Integration Issues

Issue	Common Cause	Solution
No data appearing	Credentials invalid	Verify IAM role/service account
Partial data	Permissions incomplete	Check required permissions list
Delayed data	Polling interval	CloudWatch has 5-min delay by design
Extension not running	ActiveGate issue	Check AG logs, verify assignment
Metrics but no entities	Tag configuration	Ensure resources are tagged correctly

9. Next Steps

With cloud and SaaS integrations configured:

1. **ONBRD-05: Deploying OneAgent** - Add deep application monitoring

2. **ONBRD-06: Organizing Your Environment** - Tag and segment integrated resources
3. Explore the topology in Smartscape to see cloud services
4. Create dashboards combining cloud metrics with application data

Integration Checklist

- ☐ AWS/Azure/GCP integration configured (if applicable)
 - ☐ Required cloud permissions granted
 - ☐ ActiveGate assigned for Extensions 2.0
 - ☐ Key SaaS platforms identified for integration
 - ☐ Extensions installed from Hub
 - ☐ Data verified flowing into Dynatrace
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Summary

In this notebook, you learned:

- Different integration methods (cloud, Extensions 2.0, OTel, API)
 - How to set up AWS, Azure, and GCP integrations
 - The Extensions 2.0 framework for SaaS and custom integrations
 - How to use the Dynatrace Hub to find and install integrations
 - Common SaaS integrations and their methods
 - How to verify integrations are working
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References

Cloud Integrations

- [AWS Monitoring](#)
- [Azure Monitoring](#)
- [GCP Monitoring](#)

Extensions

- [Extensions 2.0 Overview](#)
- [Extensions 2.0 Development](#)
- [Dynatrace Hub](#)

Data Ingestion

- [Metrics Ingest API](#)
- [Log Ingest](#)
- [OpenTelemetry](#)

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